1 Core Concept

1.1 What MCA Quantifies

MCA asks: if enzyme E_j is perturbed by a small amount, how much does the steady-state flux J or metabolite concentration S_i change?

$$X_0 \xrightarrow{v_1} S_1 \xrightarrow{v_2} S_2 \xrightarrow{v_3} X_1$$

Three types of coefficient capture sensitivity at different scales:

- **Elasticities** ε — local, kinetic level
- **Control coefficients** C — global, system level
- **Response coefficients** R — combined effect

1.2 Steady State

At steady state all species concentrations are constant:

$$\frac{dS_i}{dt} = 0 \quad \text{for all } i$$

A unique, stable steady state is assumed throughout.

2 Elasticity Coefficients

Elasticities are **local** properties of individual rate laws. They describe how a reaction rate responds to a change in one of its effectors, holding the enzyme level and all other variables fixed.

2.1 Scaled Elasticity

$$\boxed{\varepsilon_{ij} = \frac{\partial \ln v_i}{\partial \ln S_j} = \frac{S_j}{v_i}\frac{\partial v_i}{\partial S_j}}$$

Interpretation: a 1% change in S_j causes an ε_{ij}% change in v_i.

2.2 Unscaled Elasticity

$$\varepsilon^u_{ij} = \frac{\partial v_i}{\partial S_j}$$

Used in the Jacobian: $\mathbf{A} = \mathbf{N} \cdot \boldsymbol{\varepsilon}^u$.

2.3 Sign Convention

- $\varepsilon_{ij} > 0$ if S_j *activates* v_i (substrate)
- $\varepsilon_{ij} < 0$ if S_j *inhibits* v_i (product, inhibitor)
- $\varepsilon_{ij} = 0$ if v_i does not depend on S_j

2.4 Common Elasticity Formulas

Michaelis–Menten: $v = \dfrac{V_{\max} S}{K_m + S}$

$$\varepsilon_S = \frac{K_m}{K_m + S} \quad \in (0,\, 1)$$

Near-equilibrium ($S \ll K_m$): $\varepsilon_S \to 1$. Saturated ($S \gg K_m$): $\varepsilon_S \to 0$.

Hill kinetics: $v = \dfrac{V_{\max} S^n}{K^n + S^n}$

$$\varepsilon_S = n\,\frac{K^n}{K^n + S^n} \quad \in (0,\, n)$$

Mass action: $v = kS$

$$\varepsilon_S = 1$$

Competitive inhibition:

$$\varepsilon_I = -\frac{I/K_i}{1 + I/K_i} \quad \in (-1,\, 0)$$

Reversible Michaelis–Menten:

$$v = \frac{V_f(S/K_S) - V_r(P/K_P)}{1 + S/K_S + P/K_P}$$

$$\varepsilon_S > 0, \quad \varepsilon_P < 0$$

2.5 Haldane Relationship

For a reversible reaction at equilibrium ($v = 0$):

$$\varepsilon_S + \varepsilon_P = 0 \quad \text{and} \quad \frac{\varepsilon_S}{\varepsilon_P} = -\frac{v_f}{v_r}$$

3 Flux Control Coefficients

3.1 Definition

$$\boxed{C^J_j = \frac{\partial \ln J}{\partial \ln e_j} = \frac{e_j}{J}\frac{\partial J}{\partial e_j}}$$

C^J_j measures the fractional change in steady-state flux J caused by a fractional change in enzyme e_j, with all other enzyme levels held constant.

3.2 Alternative Form

Enzyme level e_j is proportional to $V_{\max,j}$, so equivalently:

$$C^J_j = \frac{V_{\max,j}}{J}\frac{\partial J}{\partial V_{\max,j}}$$

3.3 Summation Theorem (Flux)

$$\boxed{\sum_{j=1}^{r} C^J_j = 1}$$

Control is a shared, conserved resource: if one enzyme gains control, others must collectively lose the same amount.

3.4 Properties

- Dimensionless, scaled quantity
- Typically $0 \le C^J_j \le 1$ for unbranched pathways
- Can exceed 1 or be negative at branch points
- **Not** a fixed property of the enzyme – depends on the whole system's operating point

3.5 Intuition

- Fast enzyme (low flux limitation) $\Rightarrow$ low C^J_j
- Rate-limiting enzyme $\Rightarrow C^J_j \approx 1$
- Control is shared; there is rarely a single "rate-limiting step"

4 Concentration Control Coefficients

4.1 Definition

$$\boxed{C^{S_i}_j = \frac{\partial \ln S_i}{\partial \ln e_j} = \frac{e_j}{S_i}\frac{\partial S_i}{\partial e_j}}$$

$C^{S_i}_j$ measures how the steady-state concentration of metabolite S_i changes when enzyme e_j is perturbed.

4.2 Summation Theorem (Concentration)

$$\boxed{\sum_{j=1}^{r} C^{S_i}_j = 0 \quad \text{for all } i}$$

Scaling all enzymes by the same factor cannot change any concentration at steady state (flux rescales but S_i does

not).

4.3 Sign Interpretation

For an unbranched pathway, increasing an enzyme:

- **Decreases** concentrations of its *upstream* metabolites ($C_j^{S_i} < 0$ for i upstream of j)
- **Increases** concentrations of its *downstream* metabolites ($C_j^{S_i} > 0$ for i downstream of j)

This is the **Crossover Theorem**.

5 Connectivity Theorems

Connectivity theorems link the global control coefficients to the local elasticity coefficients.

5.1 Flux Connectivity

$$\boxed{\sum_{j=1}^{r} C_j^J \, \varepsilon_{jk} = 0 \quad \text{for all metabolites } k}$$

5.2 Concentration Connectivity

$$\boxed{\sum_{j=1}^{r} C_j^{S_i} \, \varepsilon_{jk} = -\delta_{ik} \quad \text{for all } i, k}$$

where δ_{ik} is the Kronecker delta (1 if $i = k$, else 0).

5.3 Matrix Form

Let $\mathbf{C}^J$ be the row vector of flux control coefficients and $\mathbf{C}^S$ the matrix of concentration control coefficients:

$$\mathbf{C}^J \cdot \boldsymbol{\varepsilon} = \mathbf{0}$$
$$\mathbf{C}^S \cdot \boldsymbol{\varepsilon} = -\mathbf{I}$$

Key consequence: for an n-enzyme unbranched chain (where $\boldsymbol{\varepsilon}$ is square and invertible):

$$\mathbf{C}^S = -\boldsymbol{\varepsilon}^{-1}$$

5.4 Using the Theorems

Summation + connectivity theorems provide $n + n$ equations for the n unknown flux control coefficients of an n-enzyme system. Together they fully determine $\mathbf{C}^J$ from the elasticities alone.

6 Response Coefficients

6.1 Definition

A **response coefficient** describes how a steady-state variable changes with respect to an *external* parameter p (e.g. an inhibitor concentration, an external substrate level):

$$\boxed{R_p^J = \frac{\partial \ln J}{\partial \ln p}, \qquad R_p^{S_i} = \frac{\partial \ln S_i}{\partial \ln p}}$$

6.2 Partitioned Response (Chain Rule)

If parameter p directly affects only reaction k:

$$R_p^J = C_k^J \cdot \varepsilon_{kp}$$
$$R_p^{S_i} = C_k^{S_i} \cdot \varepsilon_{kp}$$

General case (parameter affects multiple reactions):

$$R_p^J = \sum_k C_k^J \, \varepsilon_{kp}$$

6.3 Interpretation

Response = (system-level control) × (local sensitivity).

- An inhibitor with $\varepsilon_{kp} = -0.8$ applied to an enzyme with $C_k^J = 0.3$ gives $R_p^J = -0.24$: a 1% increase in inhibitor reduces flux by 0.24%.
- Even a potent inhibitor ($|\varepsilon|$ large) has little effect on flux if C_k^J is small.

7 Example: Two-Enzyme Chain

7.1 System

$$X_0 \xrightarrow{v_1} S_1 \xrightarrow{v_2} X_1$$

X_0, X_1 fixed externally; one internal species S_1. Elasticities: $\varepsilon_1 \equiv \varepsilon_{1,S_1}$ (product elasticity of v_1), $\varepsilon_2 \equiv \varepsilon_{2,S_1}$ (substrate elasticity of v_2).

Typically $\varepsilon_1 < 0$ (product inhibition), $\varepsilon_2 > 0$ (substrate activation).

7.2 Solving for C_1^J and C_2^J

Summation: $C_1^J + C_2^J = 1$

Flux connectivity ($k = S_1$): $C_1^J \varepsilon_1 + C_2^J \varepsilon_2 = 0$

$$\boxed{C_1^J = \frac{\varepsilon_2}{\varepsilon_2 - \varepsilon_1}, \qquad C_2^J = \frac{-\varepsilon_1}{\varepsilon_2 - \varepsilon_1}}$$

Since $\varepsilon_2 > 0$ and $\varepsilon_1 < 0$, both coefficients are positive and sum to 1.

7.3 Concentration Control

Summation: $C_1^{S_1} + C_2^{S_1} = 0$

Concentration connectivity: $C_1^{S_1} \varepsilon_1 + C_2^{S_1} \varepsilon_2 = -1$

$$C_1^{S_1} = \frac{1}{\varepsilon_2 - \varepsilon_1} > 0, \qquad C_2^{S_1} = \frac{-1}{\varepsilon_2 - \varepsilon_1} < 0$$

Activating E_1 (upstream) *raises* S_1; activating E_2 (downstream) *lowers* S_1.

7.4 Numeric Example

Let $\varepsilon_2 = 0.6$, $\varepsilon_1 = -0.4$.

$$C_1^J = \frac{0.6}{1.0} = 0.6, \quad C_2^J = \frac{0.4}{1.0} = 0.4$$

E_1 carries 60% of flux control; E_2 carries 40%.

8 Example: Three-Enzyme Chain

8.1 System

$$X_0 \xrightarrow{v_1} S_1 \xrightarrow{v_2} S_2 \xrightarrow{v_3} X_1$$

Relevant elasticities (v_i depends on its substrate S_{i-1} and product S_i, with external species fixed):

$$\varepsilon_{11} = \frac{\partial \ln v_1}{\partial \ln S_1} < 0 \quad \text{(product of } v_1\text{)}$$
$$\varepsilon_{21} = \frac{\partial \ln v_2}{\partial \ln S_1} > 0 \quad \text{(substrate of } v_2\text{)}$$
$$\varepsilon_{22} = \frac{\partial \ln v_2}{\partial \ln S_2} < 0 \quad \text{(product of } v_2\text{)}$$
$$\varepsilon_{32} = \frac{\partial \ln v_3}{\partial \ln S_2} > 0 \quad \text{(substrate of } v_3\text{)}$$

8.2 Connectivity Equations

For $k = S_1$: $\quad C_1^J \varepsilon_{11} + C_2^J \varepsilon_{21} = 0$
For $k = S_2$: $\quad C_2^J \varepsilon_{22} + C_3^J \varepsilon_{32} = 0$
Together with $\sum C_j^J = 1$, solving gives:

$$C_1^J \propto \frac{\varepsilon_{21}}{\varepsilon_{11}} \cdot \frac{\varepsilon_{32}}{\varepsilon_{22}}$$

$$C_2^J \propto \frac{\varepsilon_{32}}{\varepsilon_{22}}$$

$$C_3^J \propto 1$$

Normalized so that $C_1^J + C_2^J + C_3^J = 1$.

Adjacent ratio rule:

$$\frac{C_i^J}{C_{i+1}^J} = -\frac{\varepsilon_{i+1,i}}{\varepsilon_{ii}}$$

9 Example: Branched Pathway

9.1 System

$$\begin{tikzcd} X_0 \arrow[r, "v_1"] & S \arrow[r, "v_2"] \arrow[dr, "v_3"'] & X_1 \\ & & X_2 \end{tikzcd}$$

Define flux fractions: $\alpha = v_2/v_1$, $\beta = v_3/v_1$, with $\alpha+\beta = 1$ at steady state.

9.2 Branch-Point Effect

At a branch point, control coefficients can **exceed 1**:

$$C_1^{J_2} = \frac{1}{1 - \alpha\,(\varepsilon_{2S}/\varepsilon_{3S})}$$

Key result: Only enzymes *in or downstream* of a branch can control the flux ratio J_2/J_3.

Summation still holds separately for each flux: $\sum_j C_j^{J_2} = 1$ and $\sum_j C_j^{J_3} = 1$.

10 The Jacobian Connection

The unscaled elasticities link MCA to linear stability analysis.

$$\mathbf{A} = \mathbf{N} \cdot \boldsymbol{\varepsilon}^u$$

- Eigenvalues of $\mathbf{A}$ determine stability
- Time constants: $\tau_i = -1/\mathrm{Re}(\lambda_i)$
- Control coefficients relate to the DC gain: $\mathbf{C}^S \propto \mathbf{G}(0) = -\mathbf{A}^{-1}\mathbf{B}$

The **scaled** elasticities used in connectivity theorems are related to the unscaled ones by:

$$\varepsilon_{ij} = \varepsilon_{ij}^u \, \frac{S_j}{v_i}$$

11 Summary Table

Coefficient	Formula	Theorem
Elasticity ε_{ij}	$\dfrac{\partial \ln v_i}{\partial \ln S_j}$	Local; no system theorem
Flux control C_j^J	$\dfrac{\partial \ln J}{\partial \ln e_j}$	$\sum C_j^J = 1$
Conc. control $C_j^{S_i}$	$\dfrac{\partial \ln S_i}{\partial \ln e_j}$	$\sum C_j^{S_i} = 0$
Response R_p^J	$\sum_k C_k^J \, \varepsilon_{kp}$	Chain rule

Key relationships:

1. Summation + Connectivity $\Rightarrow$ solve for C from ε
2. $\mathbf{C}^S \cdot \boldsymbol{\varepsilon} = -\mathbf{I}$ (unbranched)
3. $R_p = C \cdot \varepsilon_p$ (partitioned response)
4. Control can exceed 1 at branch points

Created by Herbert M. Sauro, March 19, 2026, Based on: Heinrich & Schuster (1996); H.M. Sauro, "Systems Biology: An Introduction to Metabolic Control Analysis; Fell, Understanding the Control of Metabolism (1997)